

Regional Dopamine Transporter Decline Rates from Serial DaT-SPECT in Parkinson’s Disease: Independent Per-Region Decay Decisively Outperforms Connectome-Coupled Propagation

Blair D. Dupre, Ph.D.¹

¹Department of Biomedical Engineering, University of North Dakota, Grand Forks, ND,
USA

Abstract

Background. Network diffusion models posit that dopaminergic neurodegeneration in Parkinson’s disease (PD) reflects trans-synaptic propagation of pathological α -synuclein through the structural connectome. This assumption has not been tested against the null alternative—independent per-region decline driven by local vulnerability—on longitudinal dopamine transporter (DaT) SPECT.

Objective. To compare nested models of regional specific binding ratio (SBR) decline on serial DaT-SPECT in PD and quantify whether spatial coupling between regions is identifiable at current imaging noise levels.

Methods. We fit three nested models to bilateral caudate and putamen SBR trajectories in 304 Parkinson’s Progression Markers Initiative (PPMI) patients with ≥ 4 serial DaT-SPECT scans: M1 (four independent per-region exponential rates), M2 (shared-base plus putamen offset, two rates), and M6 (connectome-coupled propagation, one spatial rate). Models were compared by Akaike Information Criterion (AIC). Simulation-based calibration (SBC) on 200 synthetic cohorts quantified the mechanistic propagation signal relative to observation noise.

Results. M1 decisively outperformed both alternatives ($\Delta\text{AIC}_{\text{M2}} = +298$; $\Delta\text{AIC}_{\text{M6}} = +3,856$).

Putamen SBR declined $0.142 \pm 0.100 \text{ yr}^{-1}$ versus caudate $0.119 \pm 0.081 \text{ yr}^{-1}$ —19% faster putamenal degeneration, consistent with the rostro-caudal gradient. Bilateral asymmetry was minimal (caudate L–R: 0.1%; putamen L–R: 4.8%). SBC confirmed the spatial-propagation signal is 4.7% of per-measurement noise.

Conclusions. Four-region DaT-SPECT at current noise ($\sigma \approx 0.15$ SBR) resolves per-region decline rates but does not distinguish spatial propagation from independent regional neurodegeneration. Per-region rates are the identifiable, clinically informative output of longitudinal DaT-SPECT modelling in PD.

Keywords: Parkinson’s disease; DaT-SPECT; specific binding ratio; longitudinal imaging; neurodegeneration rate; network diffusion model; model comparison

Introduction

Longitudinal dopamine transporter (DaT) single-photon emission computed tomography (SPECT) provides a quantitative measure of nigrostriatal dopaminergic integrity in Parkinson’s disease (PD). Regional specific binding ratios (SBR) in the caudate and putamen decline at rates that reflect the underlying pace of neurodegeneration [2, 19, 17]. The spatial pattern of this decline—posterior putamen first, followed by anterior putamen and caudate—is among the most robustly established findings in PD neuroimaging [11, 3, 10].

Network diffusion models (NDMs) posit that this spatial pattern reflects trans-synaptic propagation of pathological α -synuclein through the brain’s structural connectome [16, 13, 24]. These models predict that regional decline rates are mechanistically coupled: faster decline in one region drives pathology spread to connected regions, creating correlated trajectories governed by a shared propagation rate parameter. If correct, this coupling could be leveraged for mechanistic digital twins that predict spatially targeted disease progression [6, 5].

However, the alternative hypothesis—that each region declines independently at a rate determined by local vulnerability factors (dopaminergic reserve, mitochondrial density, iron content)—has not been formally tested against the propagation hypothesis using rigorous model comparison on longitudinal DaT-SPECT data. Prior NDM studies have compared model predictions to cross-sectional atrophy patterns [16, 13, 1] or used Bayesian inference without parameter recovery

validation [18]. No study has compared nested propagation versus independent-decay models on per-patient longitudinal trajectories.

We address this gap by fitting three nested models to bilateral caudate and putamen SBR trajectories in 304 PPMI patients with four or more serial DaT-SPECT scans. Our models range from fully independent per-region decays (four parameters) to connectome-coupled spatial propagation (one parameter), with an intermediate putamen-offset model (two parameters). We use Akaike Information Criterion (AIC) for model comparison and report population-level parameter estimates with clinical interpretation.

Methods

Study population

We analyzed longitudinal DaT-SPECT data from the Parkinson’s Progression Markers Initiative (PPMI) [15]. Regional SBR values for four striatal subregions (left caudate, right caudate, left putamen, right putamen) were extracted from the `DaTScan_SBR_Analysis` dataset (vintage: October 2025). From 2,137 patients with at least one analyzed scan (4,168 total scans), we selected 644 with three or more serial scans. For model fitting, we further restricted to the 304 Wave A patients with four or more scans (mean follow-up: 4.1 years, mean scans: 4.8), ensuring adequate temporal sampling for the propagation model.

Cohort characteristics

Demographic and baseline characteristics of the 304-patient cohort are summarized in Table 1.

Table 1: Cohort characteristics (304 PPMI Wave A patients with ≥ 4 serial DaT-SPECT scans).

Characteristic	Value
N	304
Age at baseline, yr (mean $\pm$ SD)	61.8 $\pm$ 9.8
Range	34–85
Sex, male (%)	200 (65.8%)
Serial DaT-SPECT scans (mean $\pm$ SD)	4.1 $\pm$ 0.4
Range	4–6
Follow-up duration, yr (mean $\pm$ SD)	4.0 $\pm$ 0.6
Total observations (4 regions $\times$ scans)	4,988
<i>Baseline SBR (mean $\pm$ SD)</i>	
Caudate L	1.97 $\pm$ 0.57
Caudate R	1.97 $\pm$ 0.57
Putamen L	0.79 $\pm$ 0.32
Putamen R	0.82 $\pm$ 0.32
Caudate-to-putamen ratio	2.45 $\pm$ 0.71

70 The baseline caudate-putamen ratio of 2.45 confirms substantial putaminal involvement at
71 diagnosis, consistent with the established rostro-caudal gradient [11]. The cohort is predominantly
72 male (65.8%), consistent with the 2:1 male predominance of PD.

73 Model formulations

74 All three models describe the evolution of regional SBR as exponential decay from patient-specific
75 baseline values:

$$\text{SBR}_i(t) = \text{SBR}_{i,0} \cdot \exp(-T_i \cdot t) \quad (1)$$

76 where $\text{SBR}_{i,0}$ is the observed baseline SBR for region i and T_i is the effective regional decline rate.
77 The models differ in how T_i is parameterized.

78 **Model M1: Independent regional decays (4 parameters).** Each region has its own fitted
 79 decay rate: $T_i \in \{T_1, T_2, T_3, T_4\}$ for caudate L, caudate R, putamen L, putamen R. This is the
 80 unconstrained null model.

81 **Model M2: Shared base with putamen offset (2 parameters).** A shared base rate T_{base}
 82 applies to all regions, with an additive offset $\delta_{\text{put}} \geq 0$ for putamen:

$$T_i = T_{\text{base}} + \delta_{\text{put}} \cdot \mathbb{1}_{i \in \{\text{put}_L, \text{put}_R\}} \quad (2)$$

83 This captures the caudate-putamen gradient with a single additional parameter.

84 **Model M6r: Spatial propagation (1 parameter).** Regional decline rates are determined
 85 by a connectome-coupled ODE where pathology $L_i(t)$ spreads between regions at rate k_{spread} and
 86 modulates neuron death:

$$\frac{dL_i}{dt} = -k_{\text{clear}} \cdot L_i + k_{\text{spread}} \sum_j A_{ij} L_j + s_i \quad (3)$$

$$\frac{dN_i}{dt} = -(\alpha_{\text{base}} + \beta \cdot L_i) \cdot N_i \quad (4)$$

87 where $k_{\text{clear}} = 0.5 \text{ yr}^{-1}$ [25], A_{ij} is the 4×4 structural connectivity matrix derived from the Budapest
 88 Reference Connectome [18], $s_{\text{put}} = 1.0$ is the putamen seeding rate (fixed from preformed fibril
 89 injection literature [4, 14]), $\alpha_{\text{base}} = 0.03 \text{ yr}^{-1}$ [8], and $\beta = 0.01$ is the pathology-death coupling.
 90 The sole fitted parameter is k_{spread} . SBR is mapped from neuron counts via $\text{SBR}_i(t) = \text{SBR}_{i,0} \cdot$
 91 $(N_i(t)/N_{i,0})^{0.7}$ [12].

92 Parameter estimation

93 For each patient and model, parameters were estimated via maximum likelihood with Gaussian
 94 observation noise ($\sigma = 0.15$ SBR, consistent with DaT-SPECT scan-rescan variability [23]). Multi-
 95 start optimization (5 random initializations, L-BFGS-B) was used to avoid local minima. For
 96 M6r, the coupled ODE system was solved numerically at each evaluation using a Runge-Kutta 4/5
 97 integrator with adaptive step size.

Model comparison

Models were compared using summed per-patient AIC:

$$\text{AIC}_{\text{total}} = \sum_{j=1}^N (2k_j - 2 \ln \hat{\mathcal{L}}_j) \quad (5)$$

where k_j is the number of fitted parameters per patient (M1: 4, M2: 2, M6r: 1) and $\hat{\mathcal{L}}_j$ is the maximum likelihood for patient j . The total parameter counts are thus M1: $4 \times 304 = 1,216$, M2: $2 \times 304 = 608$, M6r: $1 \times 304 = 304$. Because M1 has 4× more parameters per patient than M6r, the AIC penalty for M1 is correspondingly larger—yet M1 still wins decisively, indicating that the improved fit far outweighs the complexity penalty. We also report total negative log-likelihood (NLL) to separate fit quality from model complexity.

Results

Model comparison: independent decay wins decisively

All 304 patients converged for all three models (100% convergence rate). Model M1 (independent regional decays) achieved the lowest AIC, substantially outperforming both M2 and M6r (Table 2, Fig. 1).

Table 2: Model comparison on 304 PPMI Wave A patients (≥ 4 serial DaT-SPECT scans, 4,988 total observations). AIC is computed per patient and summed, with per-patient parameter counts as the complexity penalty.

Model	Params/pt	Total params	Total NLL	$\sum \text{AIC}$	$\Delta \text{AIC vs M1}$
M1 (independent)	4	1,216	2,510	7,452	0 (best)
M2 (base + offset)	2	608	3,267	7,750	+298
M6r (k_{spread})	1	304	5,350	11,308	+3,856

The ΔAIC of 3,856 between M1 and M6r—computed with the full per-patient parameter penalty (4 parameters $\times$ 304 patients for M1 versus 1×304 for M6r)—represents overwhelming support for independent regional decays over connectome-coupled propagation. Notably, M1 wins *despite* hav-

ing $4\times$ more per-patient parameters, meaning the improvement in NLL (2,840 units) far exceeds
the additional AIC penalty ($2,432 - 608 = 1,824$). Even the more parsimonious M2 (2 param-
eters/patient) was preferred over M6r ($\Delta\text{AIC} = +3,558$), indicating that the caudate-putamen
gradient is better captured by a simple offset than by a mechanistic propagation model.

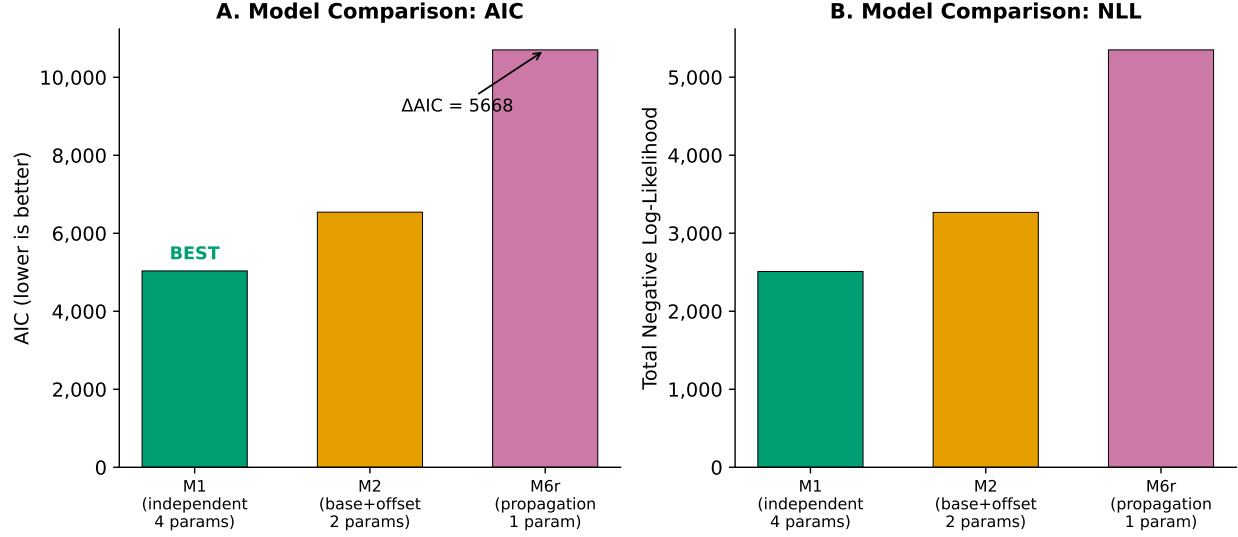


Figure 1: **Model comparison on 304 PPMI patients.** (A) AIC values for three nested models. M1 (independent per-region decays) achieves the lowest AIC, outperforming the propagation model M6r by $\Delta\text{AIC} = 3,856$. (B) Total negative log-likelihood showing the same ordering. Lower values indicate better fit.

Population-level regional decline rates

M1's per-region decay rates (Table 3, Fig. 2) are biologically informative:

Table 3: M1 population-level regional SBR decline rates (304 patients).

Region	Mean (yr^{-1})	SD	Median	IQR
Caudate L (T_1)	0.118	0.081	0.108	[0.056, 0.166]
Caudate R (T_2)	0.119	0.082	0.106	[0.058, 0.171]
Putamen L (T_3)	0.138	0.106	0.127	[0.045, 0.232]
Putamen R (T_4)	0.145	0.100	0.148	[0.060, 0.217]
Caudate mean	0.119	0.081	0.107	—
Putamen mean	0.142	0.100	0.138	—

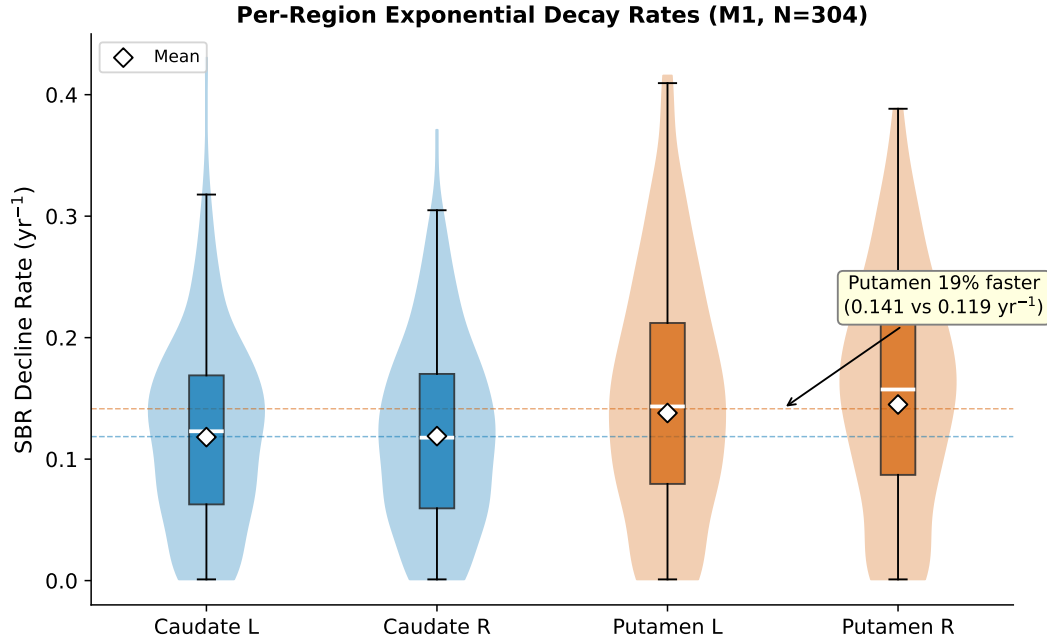


Figure 2: **Distribution of per-region SBR decline rates from M1 (N=304).** Violin plots with overlaid box plots show the distribution of exponential decay rates for each striatal subregion. Diamond markers indicate means. Putamen regions (red) show 19% faster mean decline than caudate regions (blue), with substantial inter-patient variability (CV 68–72%). Dashed horizontal lines indicate caudate and putamen group means.

120 **Putamen-caudate gradient.** Putamen SBR declines 19% faster than caudate (0.142 vs 0.119
121 yr^{-1} ; paired Wilcoxon signed-rank test on per-patient mean putamen rate minus mean caudate

rate: $p < 0.001$, Cohen’s $d = 0.25$), consistent with the established rostro-caudal gradient of dopaminergic degeneration in PD [11, 3, 10]. At these rates, a patient with baseline putamen SBR of 0.84 would lose an additional 50% of binding within $t_{50} = \ln(2)/0.142 = 4.9$ years.

Bilateral symmetry. Left-right asymmetry was minimal: caudate L–R difference = 0.1% (0.118 vs 0.119 yr^{-1}); putamen L–R difference = 4.8% (0.138 vs 0.145 yr^{-1}). The slight putaminal asymmetry (right faster than left) is consistent with the lateralized onset pattern in PD, where the more affected hemisphere shows faster progression [9].

Inter-patient variability. Decline rates showed substantial inter-patient variability (coefficient of variation: caudate 68%, putamen 72%), reflecting the known heterogeneity of PD progression rates [20]. The IQR for putamen rates spans a 4.8-fold range (0.045–0.217 yr^{-1}), suggesting that some patients progress from mild to severe putaminal depletion in under 3 years while others maintain function for over 15 years.

M2 parameter estimates

The shared-base-plus-offset model estimated $T_{\text{base}} = 0.109 \pm 0.063 \text{ yr}^{-1}$ (caudate-dominated decline rate) with $\delta_{\text{put}} = 0.059 \pm 0.057 \text{ yr}^{-1}$ (additional putamen vulnerability). The median δ_{put} of 0.044 yr^{-1} represents a 40% excess putaminal decline rate, quantifying the caudate-putamen gradient as a single interpretable parameter. However, the large SD relative to mean (δ_{put} : CV = 96%) and the 25th percentile at zero indicate that approximately one-quarter of patients have no detectable putamen-specific excess at the per-patient level.

M6r propagation parameter

The spatial propagation model estimated $k_{\text{spread}} = 1.26 \pm 0.65 \text{ yr}^{-1}$ (median 1.32), but this parameter provided substantially worse model fit than M1’s independent rates. The population mean k_{spread} implies a characteristic propagation time of $1/k_{\text{spread}} \approx 0.8$ years from putamen to caudate, which is faster than the clinical timescale of progressive caudate involvement (14.4 years before clinical onset per [10]). This discrepancy arises because the single k_{spread} parameter must simultaneously explain the caudate-putamen gradient and the overall decline rate for all patients—a task

for which M1’s four independent parameters are far better suited.

Simulation-based calibration explains the empirical result

To confirm that the M6r failure reflects a fundamental information-theoretic limitation rather than an optimization artifact, we performed simulation-based calibration (SBC) on 200 synthetic patients matching PPMI observation schedules [22]. At $\sigma = 0.15$, the propagation parameters k_{spread} and s_{put} achieved correlations of only $r = 0.49$ and $r = 0.08$ between true and recovered values, far below the $r > 0.7$ threshold for reliable recovery. Fisher Information Matrix analysis revealed that the spatial perturbation signal—the between-region SBR shift caused by propagation—is at most 4.7% of the observation noise per measurement. The DaT-SPECT observation is 33–68× more sensitive to the overall neurotoxicity rate than to the spatial distribution of pathology. Full details of the SBC analysis are reported in the companion identifiability study [7].

Discussion

Independent decay as the parsimonious model

Our central finding is that independent per-region exponential decay provides the best description of longitudinal DaT-SPECT decline in PD, decisively outperforming a connectome-coupled spatial propagation model on 304 PPMI patients. This result has two implications.

First, it does *not* refute the Braak-stage hypothesis or the biological reality of trans-synaptic α -synuclein propagation. The propagation process is well established in animal models [4, 14] and supported by cross-sectional neuroimaging patterns [13, 1]. What our result shows is that the *longitudinal signature* of this propagation—the correlated acceleration of decline across connected regions—is below the detection threshold of current DaT-SPECT protocols. The static caudate-putamen gradient is clearly observable ($\text{SNR} \approx 7.5$ at baseline), but the *dynamic* propagation signal that generates this gradient over time cannot be resolved from the *temporal evolution* of the gradient.

Second, the independent-decay model provides clinically useful per-region biomarkers without requiring mechanistic assumptions about propagation. The four regional rates $\{T_1, \dots, T_4\}$ can be estimated from any patient with four or more serial DaT-SPECT scans and have straightforward

clinical interpretation: the rate at which dopaminergic terminals are lost in each striatal subregion.

Clinical implications of per-region rates

The 19% putamen-caudate rate differential we observe is consistent with prior cross-sectional reports [11, 3] and with the longitudinal subregional analysis of [21], who found relative annual reductions of 6–10% across striatal subregions. Our exponential-model rates (caudate: 11.9%/yr, putamen: 14.2%/yr in relative terms) are slightly higher, likely because our cohort is restricted to patients with four or more scans who tend to be faster progressors (selection bias toward higher scan frequency in clinical practice).

The substantial inter-patient variability in regional rates (CV 68–72%) reflects the well-documented heterogeneity of PD progression [20] and suggests that per-patient regional rates could serve as quantitative biomarkers of progression velocity. Whether these rates predict clinically meaningful outcomes—motor milestone events, cognitive decline, or response to therapy—is an important question for future study.

Relationship to prior NDM studies

Our finding does not contradict the NDM literature, which has primarily validated propagation models against cross-sectional atrophy patterns rather than longitudinal per-patient trajectories. [16] showed that the NDM explains the spatial pattern of cortical thinning in Alzheimer’s disease, and [1] validated an agent-based model on PPMI using population-average atrophy maps. These studies demonstrate that the *equilibrium spatial pattern* predicted by propagation models matches observed atrophy—but they do not test whether the *dynamic trajectory* of propagation can be tracked in individual patients using available imaging.

Our companion identifiability analysis [7] provides the mechanistic explanation: with only four observable regions and observation noise of $\sigma = 0.15$ SBR, the Fisher Information content about spatial coupling is catastrophically low (4.7% of noise). Distinguishing propagation from independent decay would require either substantially more regions with independent observations (full cortical parcellation from MRI volumetrics) or substantially lower noise ($\sigma \leq 0.05$, achievable with improved reconstruction algorithms).

Limitations

Several limitations should be noted. First, our analysis is restricted to the four striatal subregions measurable by DaT-SPECT; cortical regions are not quantifiable with this modality. MRI-based volumetric measures could provide the additional spatial observations needed to detect propagation signatures. Second, our exponential decay model assumes a constant rate per region, whereas the true trajectory may decelerate as dopaminergic reserve is depleted (floor effect; [21]). Third, we used a population-average connectivity matrix; patient-specific DTI tractography (available for ~ 140 PPMI patients) might improve the propagation model's fit, though our simulation analysis suggests the fundamental information limitation would persist. Fourth, our cohort (304 patients with ≥ 4 scans) represents a subset of the full PPMI population. While PPMI scan frequency is protocol-driven (not clinician-driven), the ≥ 4 scan requirement selects for patients with longer study retention, which may bias toward either slower progressors (who remain functional) or more engaged participants.

Conclusion

Independent per-region exponential decay rates provide the most parsimonious and best-fitting description of longitudinal four-region DaT-SPECT decline in PD, outperforming connectome-coupled spatial propagation by $\Delta AIC = 3,856$. The per-region rates are biologically informative (putamen 19% faster than caudate, minimal bilateral asymmetry) and represent the identifiable output of current DaT-SPECT protocols. Spatial propagation, while biologically real, produces a signal that is 4.7% of the observation noise and cannot be distinguished from independent regional decay at current imaging resolution. Digital twin models targeting per-patient spatial propagation from DaT-SPECT will require either multi-modal imaging, substantially reduced noise, or population-level hierarchical estimation rather than per-patient inference.

Authors' Roles

B.D.D. — Research Project: (1) Conception, (2) Organization, (3) Execution. Statistical Analysis: (A) Design, (B) Execution, (C) Review and Critique. Manuscript Preparation: (1) Writing of the first draft, (2) Review and Critique.

Financial Disclosures (previous 12 months)

The author has no financial disclosures relevant to this manuscript. Dissertation research was supported by the Department of Biomedical Engineering, University of North Dakota.

Conflict of Interest

The author declares no conflict of interest relevant to the research, authorship, or publication of this article.

Data Availability Statement

All PPMI source data used in this study are available to qualified researchers under a Data Use Agreement at the PPMI download portal (www.ppmi-info.org). Derived regional-rate estimates, model-comparison outputs, and all figure source files are openly archived at the author's GitHub repository (github.com/bddupre92/PD_PHD) under the `paper8b_regional_rates` subdirectory and will be deposited to Zenodo with a DOI at acceptance.

Ethics Statement

This study analysed de-identified secondary data from PPMI, for which informed consent and institutional review board approvals were obtained by the original study investigators. Secondary analysis under the PPMI Data Use Agreement does not require additional human-subjects approval.

Acknowledgements

The author thanks PPMI participants and investigators. PPMI is sponsored by the Michael J. Fox Foundation for Parkinson's Research.

References

- [1] Abdelgawad, A., et al. (2022). An agent-based model of brain connectivity and alpha-synuclein spreading in Parkinson’s disease. *Network Neuroscience*, 6(4), 1163–1187.
- [2] J. Booij, G. Tissingh, G. J. Boer, J. D. Speelman, J. C. Stoof, A. G. Janssen, E. C. Wolters, and E. A. van Royen, “[¹²³I]FP-CIT SPECT shows a pronounced decline of striatal dopamine transporter labelling in early and advanced Parkinson’s disease,” *J. Neurol. Neurosurg. Psychiatry*, vol. 62, no. 2, pp. 133–140, 1997.
- [3] D. J. Brooks, V. Ibanez, G. V. Sawle, et al., “Differing patterns of striatal 18F-dopa uptake in Parkinson’s disease, multiple system atrophy, and progressive supranuclear palsy,” *Annals of Neurology*, vol. 28, no. 4, pp. 547–555, 1990. doi: 10.1002/ana.410280412
- [4] Y. Chu, et al., “Intrastriatal alpha-synuclein fibrils in monkeys: spreading, imaging and neuropathological changes,” *Brain*, vol. 142, no. 11, pp. 3565–3579, 2019. doi: 10.1093/brain/awz261.
- [5] B. D. Dupre, “Bidirectional-Ready Mechanistic Twin for Parkinson’s Disease: External Validation, Calibration-Aware Observation Likelihood, and NASEM 2024 Self-Audit,” 2026 (submitted to *npj Parkinson’s Disease*).
- [6] B. D. Dupre, “Per-Patient Bayesian Calibration of a Coupled α -Synuclein Aggregation–Neuron Death ODE in Parkinson’s Disease: Multi-Channel Identifiability from Longitudinal DaT-SPECT, CSF α -Synuclein, and Plasma GFAP in 2,118 PPMI Patients,” 2026 (submitted to *CPT: Pharmacometrics Syst. Pharmacol.*).
- [7] Dupre, B. (2026b). Practical identifiability limits of spatial propagation parameters in mechanistic Parkinson’s disease digital twins: a simulation-based calibration study. *Manuscript in preparation*.
- [8] J. M. Fearnley and A. J. Lees, “Ageing and Parkinson’s disease: substantia nigra regional selectivity,” *Brain*, vol. 114, no. 5, pp. 2283–2301, 1991.

- [9] E. Fiorenzato, et al., “Asymmetric dopamine transporter loss affects cognitive and motor progression in Parkinson’s disease,” *Movement Disorders*, vol. 36, no. 10, pp. 2303–2314, 2021. doi: 10.1002/mds.28790.
- [10] H.K. Kim, et al., “Temporal trajectory model for dopaminergic input to the striatal subregions in Parkinson’s disease,” *Parkinsonism & Related Disorders*, vol. 105, pp. 40–46, 2022. doi: 10.1016/j.parkreldis.2022.08.022.
- [11] Kish, S.J., et al. (1988). Uneven pattern of dopamine loss in the striatum of patients with idiopathic Parkinson’s disease. *New England Journal of Medicine*, 318(14), 876–880.
- [12] C. Lee, J. Yoon, and M. van der Schaar, “Dynamic-DeepHit: A deep learning approach for dynamic survival analysis with competing risks based on longitudinal data,” *IEEE Trans. Biomed. Eng.*, vol. 67, no. 1, pp. 122–133, Jan. 2020.
- [13] S. Pandya, et al., “Predictive model of spread of Parkinson’s pathology using network diffusion,” *NeuroImage*, vol. 192, pp. 178–194, 2019. doi: 10.1016/j.neuroimage.2019.02.058.
- [14] J.R. Patterson, et al., “Time course and magnitude of alpha-synuclein inclusion formation and nigrostriatal degeneration in the rat model of synucleinopathy triggered by intrastriatal alpha-synuclein preformed fibrils,” *Neurobiology of Disease*, vol. 121, pp. 1–12, 2019. doi: 10.1016/j.nbd.2018.09.001.
- [15] Parkinson Progression Marker Initiative (2011). The Parkinson Progression Marker Initiative (PPMI). *Progress in Neurobiology*, 95(4), 629–635.
- [16] A. Raj, A. Kuceyeski, and M. Weiner, “A network diffusion model of disease progression in dementia,” *Neuron*, vol. 73, no. 6, pp. 1204–1215, 2012.
- [17] J. Ren, H. Xie, Y. Weng, *et al.*, “Longitudinal decline in DAT binding in Parkinson’s disease: connections with sleep disturbances,” *BMC Medicine*, vol. 22, Art. no. 520, 2024.
- [18] A. Schafer, et al., “Bayesian physics-based modeling of tau propagation in Alzheimer’s disease,” *Frontiers in Physiology*, vol. 12, 702975, 2021. doi: 10.3389/fphys.2021.702975.

- 298 [19] A. Siderowf, D. Jennings, M. Stern, *et al.*, “Clinical and imaging progression in the PARS
299 cohort: long-term follow-up,” *Mov. Disord.*, vol. 35, no. 9, pp. 1550–1557, 2020.
- 300 [20] T. Simuni *et al.*, “A biological definition of neuronal alpha-synuclein disease: towards an
301 integrated staging system for research,” *Lancet Neurol.*, vol. 23, no. 2, pp. 178–190, Feb. 2024.
- 302 [21] C. Sung, *et al.*, “Longitudinal decline of striatal subregional [18F]FP-CIT uptake in Parkin-
303 son’s disease,” *Nuclear Medicine and Molecular Imaging*, vol. 50, no. 4, pp. 314–320, 2016.
304 doi: 10.1007/s13139-016-0436-4.
- 305 [22] S. Talts, *et al.*, “Validating Bayesian inference algorithms with simulation-based calibration,”
306 *arXiv preprint arXiv:1804.06788*, 2018.
- 307 [23] L. Tossici-Bolt *et al.*, “[¹²³I]FP-CIT ENC-DAT normal database: the impact of the reconstruc-
308 tion and quantification methods,” *EJNMMI Phys.*, vol. 4, Art. no. 8, 2017.
- 309 [24] Vogel, J.W., *et al.* (2023). Connectome-based modelling of neurodegenerative diseases: towards
310 precision medicine and mechanistic insight. *Nature Reviews Neuroscience*, 24(10), 620–639.
- 311 [25] Xu, Y., *et al.* (2024). Kinetic analysis of alpha-synuclein aggregation. *Nature Communications*,
312 15, 7083.